

Climate Risk Assessment of Munshiganj Municipality

Vulnerability Risk Using Secondary Sources

A comprehensive climate risk assessment was conducted for Munshiganj Municipality using secondary spatial datasets, satellite imagery, and GIS-based analysis. The assessment focused on identifying areas exposed to climate-related risks by analyzing key environmental indicators, including land use and land cover change, land surface temperature, rainfall variation, and flood vulnerability.

The assessment was conducted using Google Earth Engine (GEE) and GIS-based spatial analysis techniques. Satellite imagery and environmental datasets were collected from globally recognized sources, including USGS Landsat, Copernicus Sentinel-2, CHIRPS rainfall dataset, NASA SRTM Digital Elevation Model (DEM), and OpenLandMap soil dataset.

The analysis provides a spatial understanding of climate exposure and supports the identification of priority areas for climate adaptation interventions within Munshiganj Municipality.

Overlay Analysis of Risk Assessment

Climate vulnerability in urban areas is influenced by multiple interacting environmental factors. Therefore, an overlay-based assessment approach was adopted to identify areas experiencing higher exposure to climate risks.

Four major environmental indicators were considered:

- Land Use and Land Cover (LULC)
- Land Surface Temperature (LST)
- Rainfall Pattern
- Flood Vulnerability

These indicators provide a comprehensive understanding of urban expansion, environmental degradation, rainfall variability, and flood exposure within the municipality.

Land Cover Classification and Land Use Change

Land Use and Land Cover (LULC) classification was conducted to understand changes in natural and built environments of Munshiganj Municipality. Changes in vegetation cover, water bodies, built-up areas, and bare land influence climate vulnerability by affecting surface temperature, drainage conditions, and urban heat exposure.

Using GIS-based satellite image analysis, LULC maps were prepared for 1995 and 2025 to identify major land transformation patterns within the municipality.

Year	Dataset	Source	Resolution
1995	Landsat 5 TM Surface Reflectance	USGS Landsat Archive	30 m
2025	Sentinel-2 MSI Surface Reflectance	Copernicus Sentinel Programme	10 m

Land Surface Temperature Map

Increasing temperature is one of the major impacts of climate change, creating heat stress and affecting urban comfort. Land Surface Temperature (LST) mapping was conducted to identify spatial variation of surface temperature within Munshiganj Municipality.

Satellite-derived thermal data were used to prepare LST maps for different periods and identify areas exposed to higher heat intensity.

Year	Dataset	Source
1995	Landsat 5 Thermal Band	USGS Landsat Archive
2025	Landsat 8 Thermal Band	USGS Landsat Archive

Rainfall Map

Changes in rainfall patterns and increasing intensity of rainfall events contribute to flooding and waterlogging problems in urban areas. Therefore, rainfall analysis was conducted to understand spatial variation of rainfall conditions in Munshiganj Municipality.

Dataset	Source	Period
CHIRPS Daily Rainfall	Climate Hazards Group (CHG), UCSB	2015–2025

Flood Risk Map

Flooding is one of the major climate hazards affecting Munshiganj due to its low-lying floodplain characteristics and proximity to major river systems. A GIS-based flood risk assessment was conducted by integrating topographic, hydrological, climatic, and land surface characteristics.

Parameter	Source
Elevation (DEM)	NASA SRTM DEM
Slope	Derived from DEM
Rainfall	CHIRPS Dataset
Land Use/Land Cover	Satellite Image Classification
Distance from River	River Network Data
Soil Condition	OpenLandMap Soil Dataset

Data were analyzed using Google Earth Engine. Areas with lower elevation, gentle slope, higher rainfall exposure, and close proximity to water bodies were considered more vulnerable to flooding. A weighted overlay approach was applied to combine different flood-related indicators and prepare the flood risk map.

Flood Risk Index (FVI); Each parameter was reclassified into three risk levels: Low = 1, Moderate = 2, High = 3

The Flood Risk Index was calculated as:

$$FVI = 0.25 \text{ (DEM)} + 0.20 \text{ (Distance from River)} + 0.20 \text{ (Rainfall)} + 0.15 \text{ (Slope)} + 0.12 \text{ (LULC)} + 0.08 \text{ (Soil)}$$

Climate Risk Map Preparation Using Satellite Data

To identify areas facing multiple climate stresses, an overlay analysis was conducted by integrating LULC, LST, rainfall, and flood vulnerability information. This approach helped identify climate-sensitive areas where multiple risks overlap, such as: high surface temperature, reduced vegetation coverage, intense rainfall exposure, flood susceptibility. The findings provide a scientific basis for prioritizing climate adaptation interventions, including flood management, drainage improvement, heat reduction, and green infrastructure development.

Results and Discussions

Land Use and Land Cover Change

The LULC maps show a clear expansion of built-up areas between 1995 and 2025. At the municipal level, built-up land increased from 5.41% to 8.82%, representing a 63.6% increase, while vegetation, water bodies and bare land declined slightly.

At the ward level, the strongest concentration and expansion of built-up land is visible in Wards 1, 2, 3 and 4, which form the established urban core. Ward 2 and Ward 3 show particularly dense development along the major road corridors, while built-up areas have also become more scattered in Wards 5 and 9 by 2025. In contrast, Wards 7, 8 and 9 remain comparatively open and agriculture-dominated, although scattered development is becoming visible. Ward 6 retains relatively extensive vegetation, while vegetation is also concentrated along canals, homesteads and field boundaries in Wards 7–9. The pattern reflects the gradual outward expansion of the municipality from the established western and central urban areas toward the agricultural eastern and southern wards. The existing land-use survey similarly identifies Wards 1–4 as the urban core, Wards 5–6 as transitional areas and Wards 7–9 as predominantly agricultural peri-urban areas.

Table 1: Change in Land Cover Classes from 1995 to 2025

LULC Class	% 1995	% 2025	Rate of Change (%)	Trend
Water body	3.48%	3.40%	-1.9%	▼ Dec.
Vegetation	29.96%	28.99%	-2.8%	▼ Dec.
Built-up	5.41%	8.82%	+63.6%	▲ Inc.
Bare land	61.15%	58.79%	-3.4%	▼ Dec.

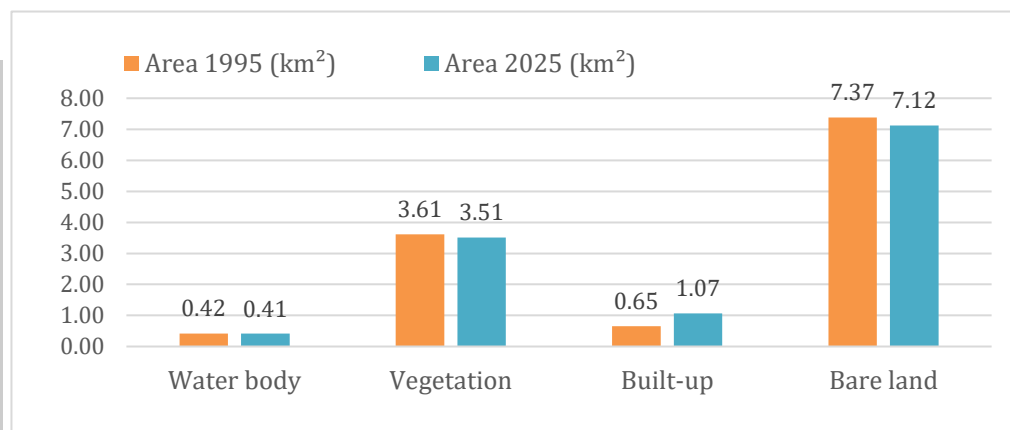
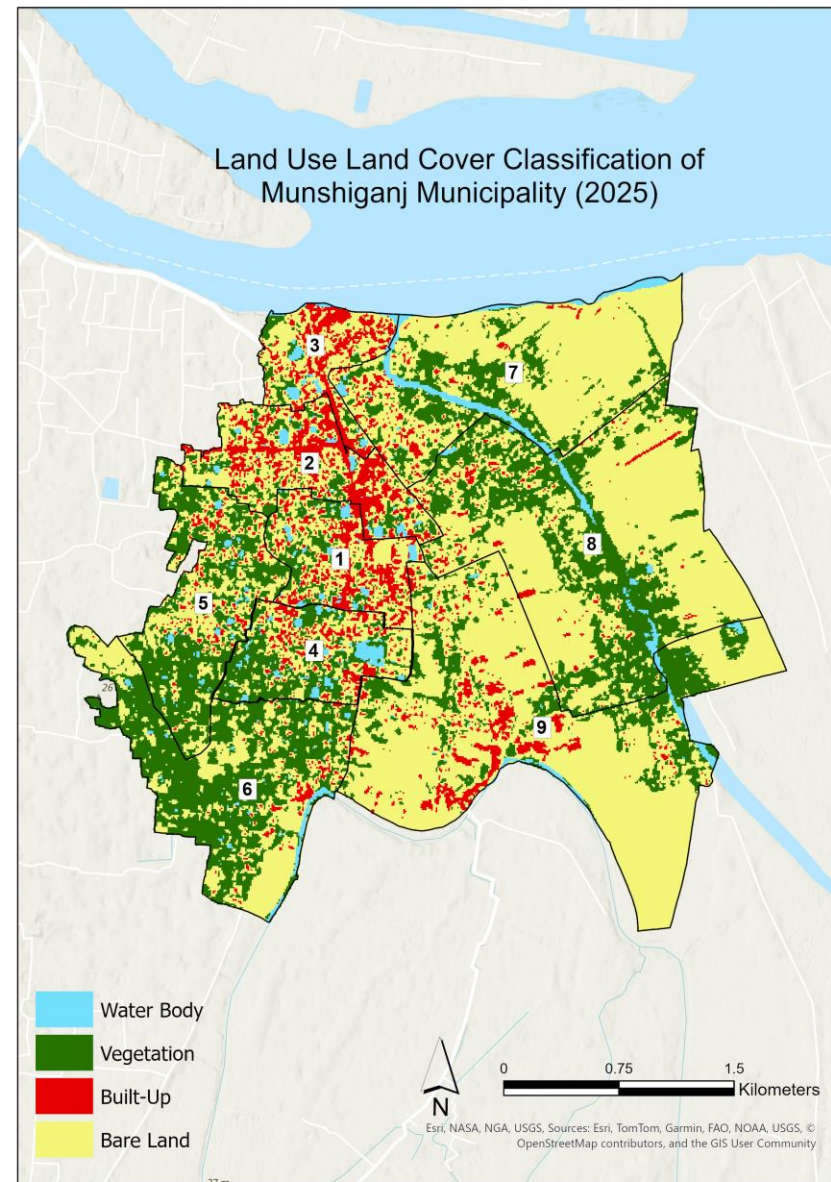
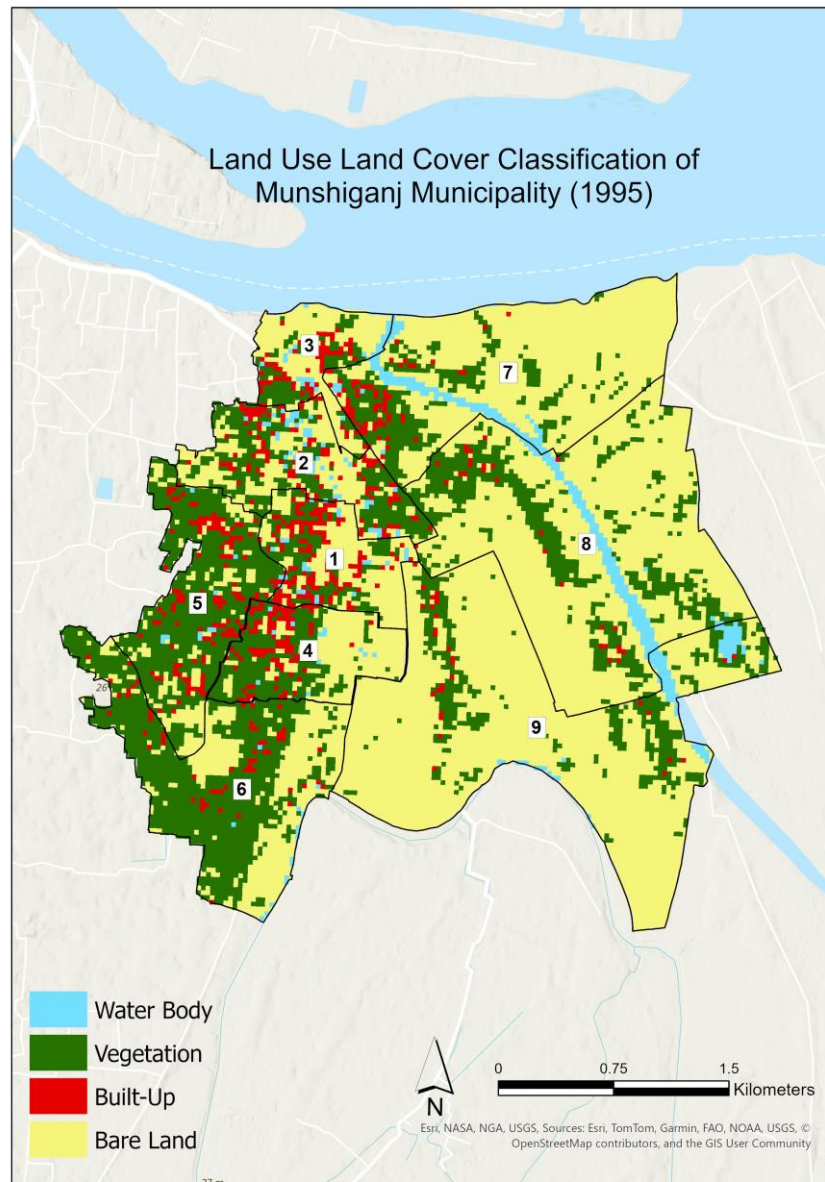


Figure 1: Change in Land Cover Classes from 1995 to 2025



Map 1: Land Use Land Cover Change of Munshiganj Municipality from 1995 to 2025

Land Surface Temperature (LST)

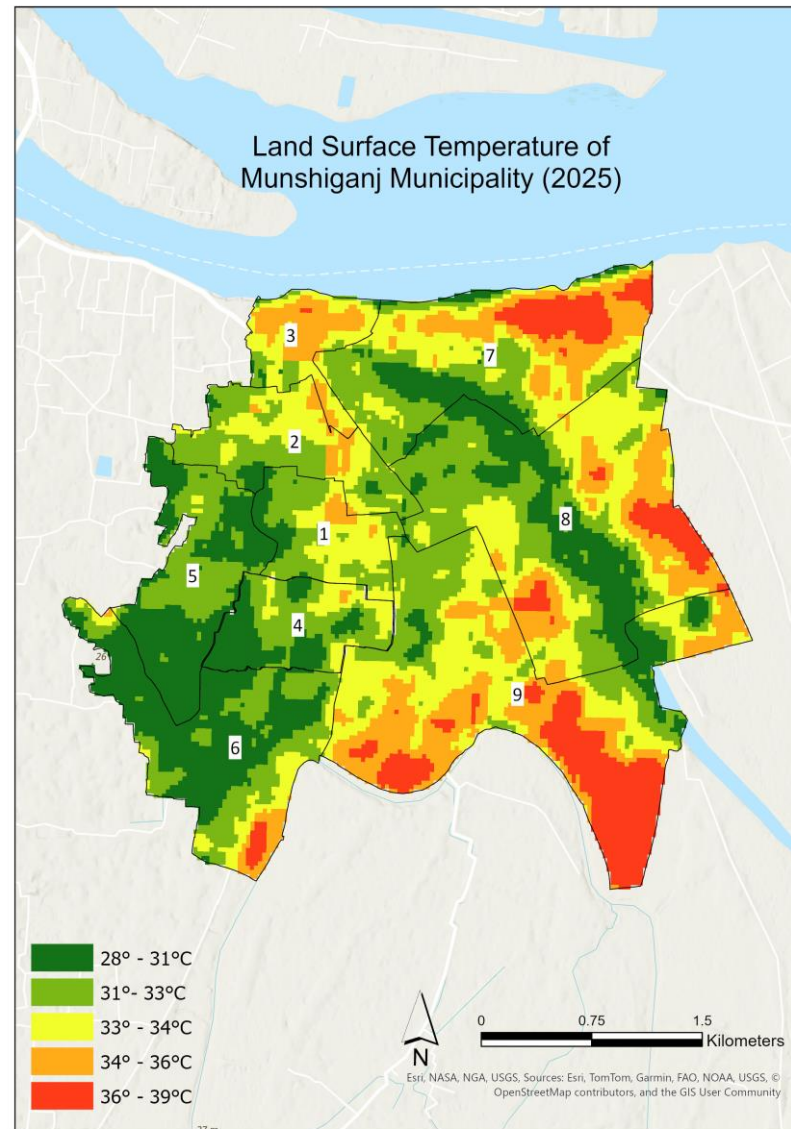
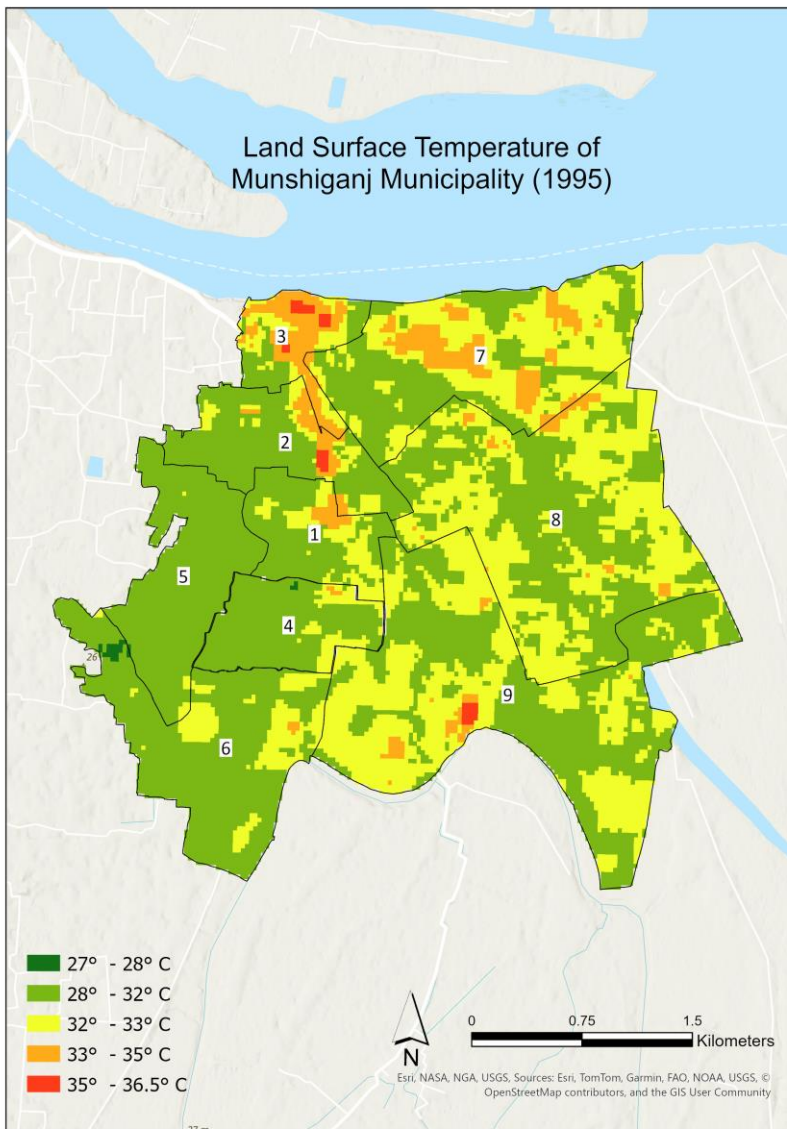
The LST analysis shows a clear increase in surface temperature between 1995 and 2025. The highest temperature class increased from approximately 35–36.5°C in 1995 to 36–39°C in 2025, indicating an increase of about 3°C in the peak LST range.

In 2025, the highest LST is concentrated mainly in Wards 7, 8 and 9, while localized high-temperature areas are also visible in Ward 3. In contrast, Wards 5 and 6 remain comparatively cooler, with large areas falling within approximately 28–33°C. Wards 1, 2 and 4 generally show moderate surface temperatures. The pattern indicates that high LST in Munshiganj is not controlled by built-up density alone.

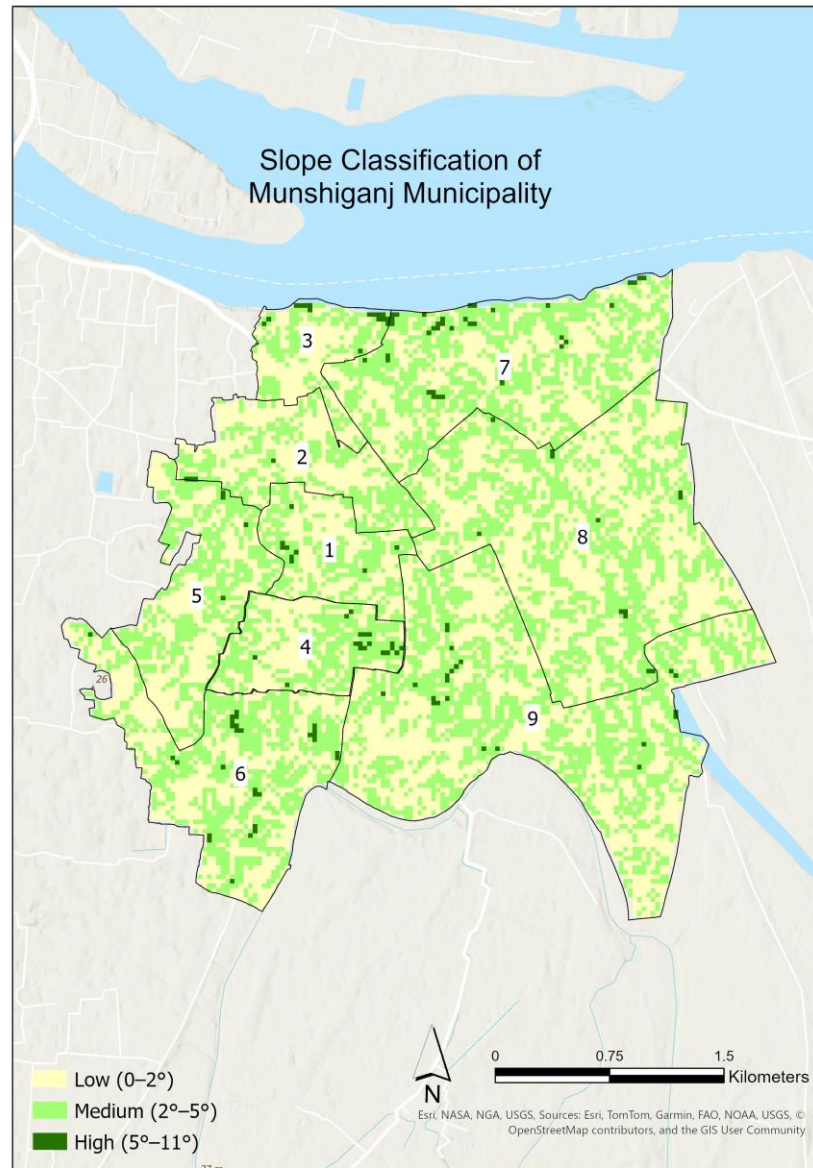
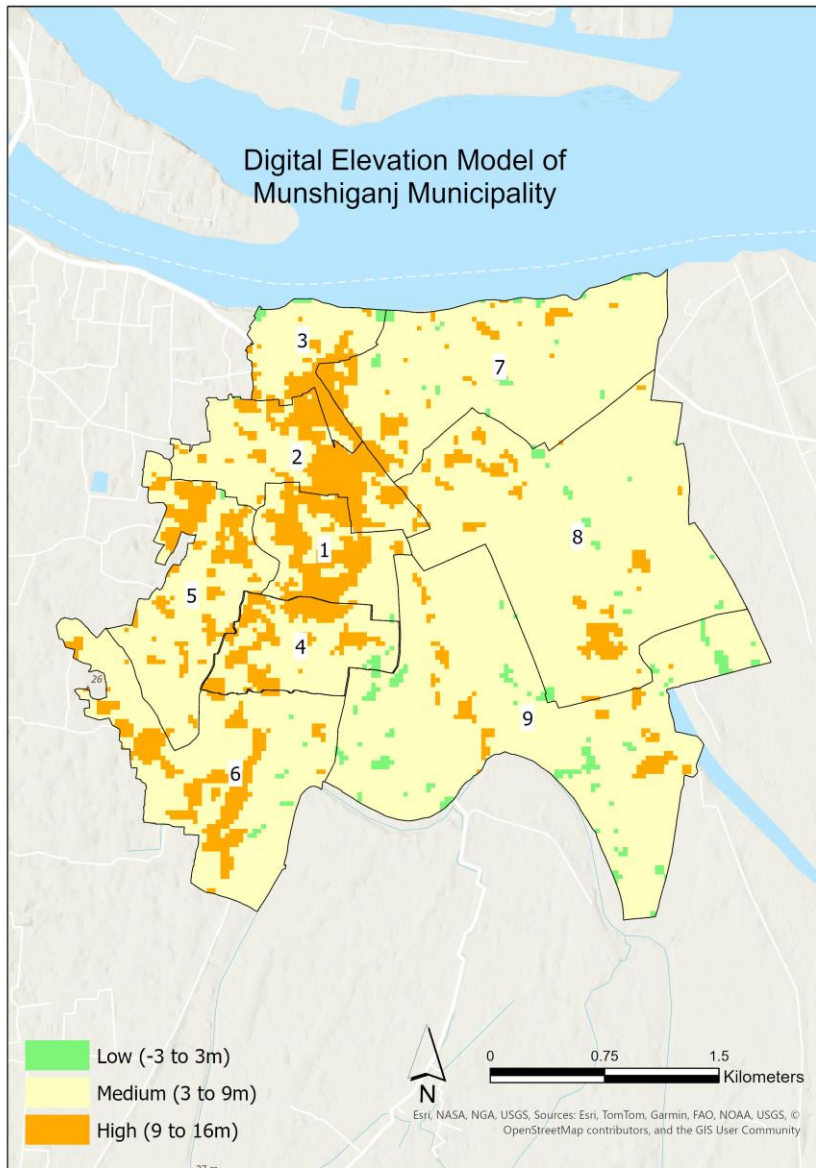
Although Wards 7–9 are less densely built-up and are located closer to rivers, they are predominantly agricultural. Munshiganj Municipality follows a diverse seasonal cropping pattern dominated by vegetables, potato, Aman rice and Boro rice. April represents an important transition period, when potato and Boro fields are often harvested or recently harvested and some vegetable plots have relatively low canopy cover. Consequently, extensive agricultural areas may temporarily consist of exposed, dry or stubbled soil, particularly in the agriculture-dominated Wards 7–9. These seasonal surface conditions can produce relatively high land surface temperatures despite the lower built-up density of these wards.

The cooling influence of rivers also appears to be relatively localized. In Ward 8, for example, cooler conditions are visible along the canal and vegetated corridor, while higher temperatures occur farther away over open agricultural surfaces. Similarly, the hotter zones in Ward 7 are concentrated mainly toward the northern and northeastern agricultural areas, whereas Ward 9 shows extensive high-temperature patches across its southern and eastern agricultural land.

Overall, the LST pattern suggests that the high temperatures observed in Wards 7–9 are more strongly associated with seasonal agricultural surface conditions and exposed soil than with urban built-up concentration. Conversely, the relatively lower LST in Wards 5 and 6 corresponds with their greater vegetation cover, which provides more shading and evapotranspirative cooling.



Map 2: Land Surface Temperature (LST) Comparison between 1995 and 2025



Rainfall Variation

Rainfall varies considerably from year to year (Figure 15), with distinct wet and dry periods and no uniform long-term trend. The sharp fluctuations indicate increasing importance of managing both intense rainfall and seasonal waterlogging risks.

Across Munshiganj Municipality, rainfall range varies from approximately 2,166 to 2,197 mm. The lowest rainfall class, around 2,166–2,175 mm, is concentrated mainly in Wards 5 and 6, with parts of Wards 1–4 also falling within the lower range. Rainfall gradually increases toward the east. Wards 7, 8 and 9 receive comparatively higher rainfall, with Ward 8 and the eastern parts of Wards 7 and 9 showing the highest values of approximately 2,185–2,197 mm.

Although the absolute difference across the municipality is not large, this spatial gradient becomes important when combined with low elevation, proximity to rivers and poor drainage.

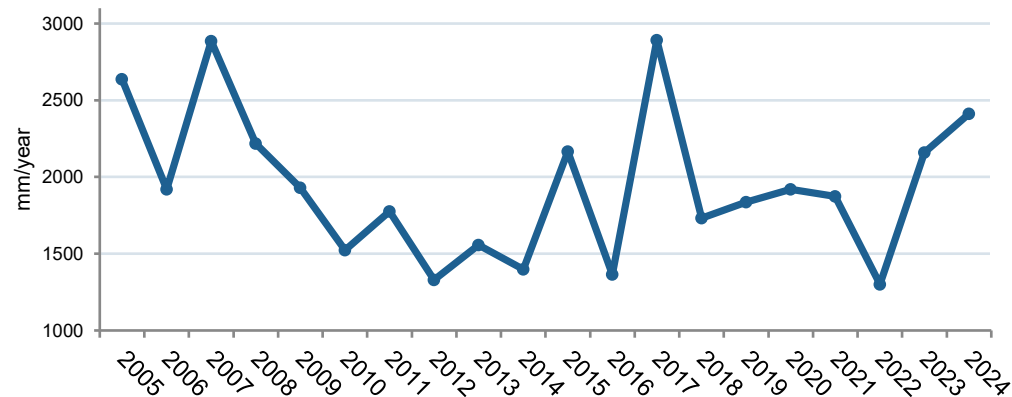
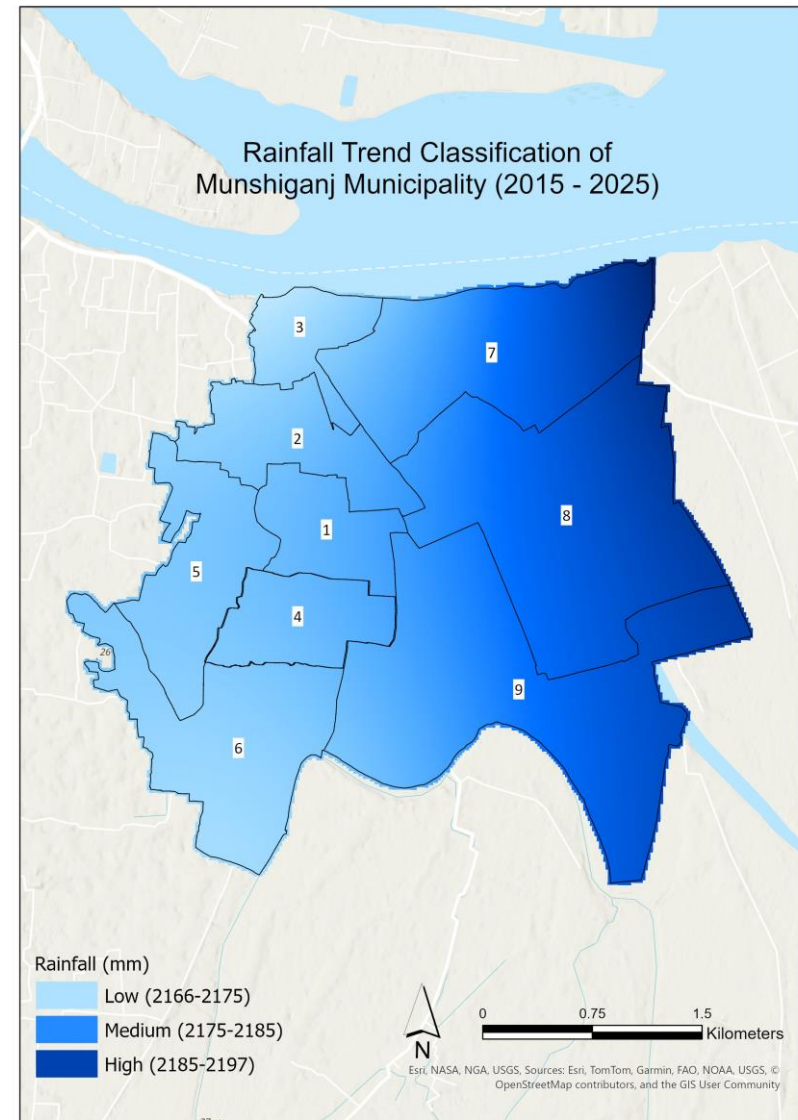


Figure 1: Annual Rainfall Trend from 2005 to 2024

Source: BMD



Map 3: 10 Years Rainfall Trend (2015 - 2025) of Munshiganj Municipality

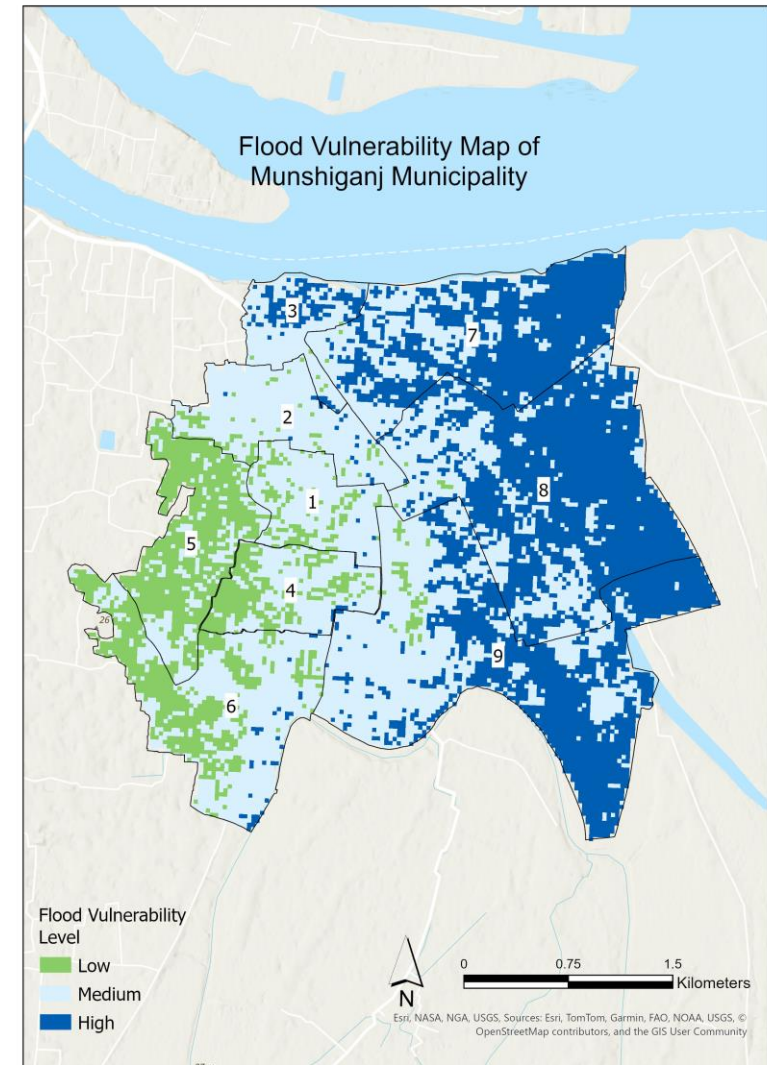
Flood Risk

The Flood Vulnerability Map of Munshiganj Municipality shows a pronounced west-to-east gradient in flood exposure across the municipality.

The eastern and southeastern wards — Ward 7, Ward 8 and Ward 9 — are classified as high flood vulnerability zones. This pattern is consistent with the physical setting of these wards: they occupy the low-lying, peri-urban floodplain closest to the Meghna River, fall within the higher end of the municipality's rainfall gradient (approximately 2,185–2,197 mm), and retain the lowest built-up density and highest share of exposed agricultural land, all of which reduce natural resistance to inundation.

The central wards — Ward 1, Ward 2, Ward 3 (southern portion) and Ward 4 — fall predominantly into the medium vulnerability class. These wards form the older, more built-up urban core, where higher elevation and denser road/drainage infrastructure moderate direct river-flood exposure, though localized medium- and even isolated high-risk pockets remain, particularly along Ward 3's eastern boundary near the canal branch connecting to the Dhaleshwari River.

The western wards — Ward 5 and Ward 6 — show the lowest flood vulnerability in the municipality, with low-risk pixels dominating both wards. This corresponds to their comparatively higher ground elevation, lower rainfall exposure (2,166–2,175 mm), and greater distance from the Meghna and Dhaleshwari river systems.



Map 4: Flood Vulnerability Map of Munshiganj Municipality

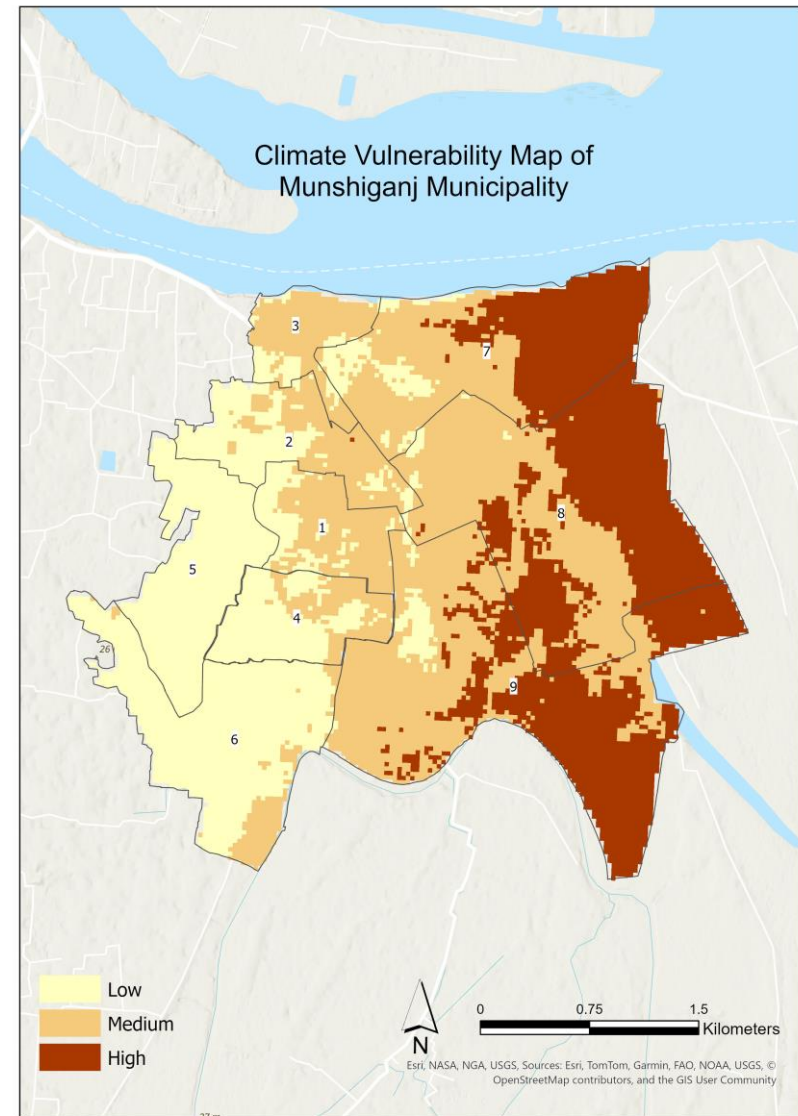
Climate Vulnerability Assessment

To capture the combined effect of multiple climate stressors, an overlay analysis integrating Land Use/Land Cover, Land Surface Temperature, rainfall pattern and the Flood Vulnerability Index was conducted for Munshiganj Municipality.

The result shows a similarly clear spatial gradient. Ward 7, Ward 8 and the greater part of Ward 9 are classified as high climate vulnerability, forming a large, contiguous high-risk zone along the eastern and southeastern edge of the municipality, coinciding with the Meghna River corridor. This is the combined product of the highest flood exposure, comparatively higher pre-monsoon land surface temperature (linked to exposed, harvested or fallow agricultural surfaces, as discussed in Section 4.1.2), and the upper end of the municipal rainfall range identified in these wards.

Ward 1, Ward 2, Ward 3 and Ward 4 — the established urban core and its immediate transition zone — fall into the medium vulnerability category. Here, moderate LST, medium flood risk and mid-range rainfall combine with denser built-up land use to produce an intermediate overlay score.

Ward 5 and Ward 6, in the west, register the lowest overlay scores in the municipality, consistent with their lower flood exposure, higher retained vegetation cover, and comparatively lower LST and rainfall values documented in the preceding subsections.



Map 5: Climate Vulnerability Map of Munshiganj Municipality Overlaying Satellite Data